PCB Layout for EMI/EMC Compliance

Proven Best Practice layout tips

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This guide-line made to document the prove of concept PCB layout impact on EMI/ EMC test especially for the DV/ DP test where the most failing test we are facing

This is not a final product this pcb not intended also for assembly in and body the main goal is prove of concept ideas ONLY

Scope of the document:

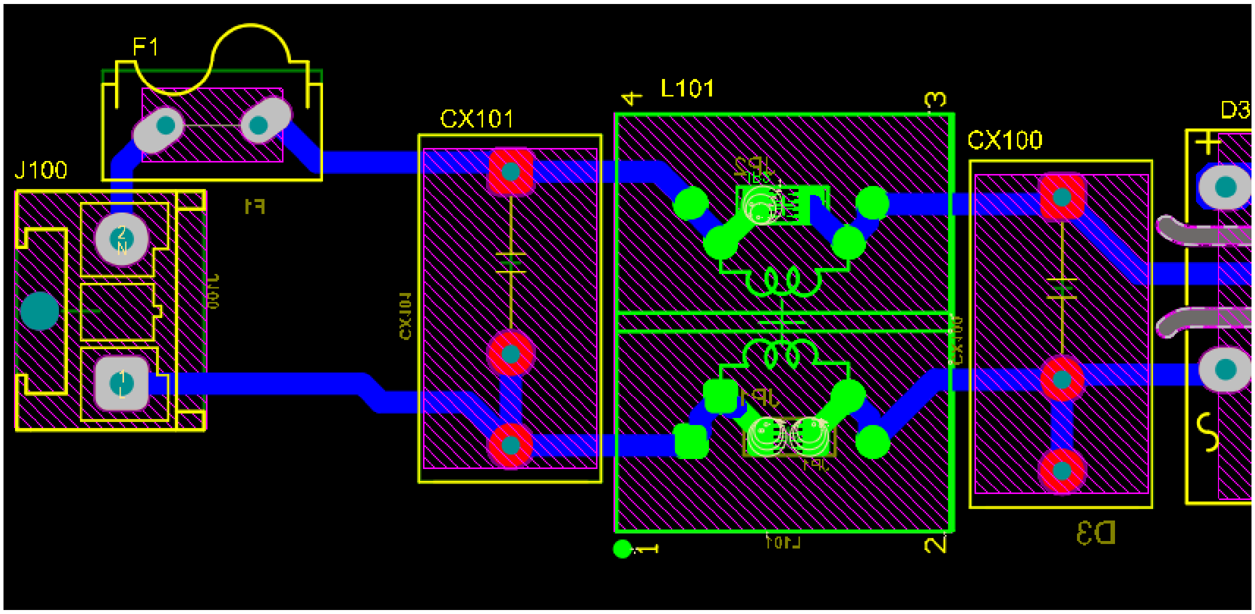
Verify the pcb layout impact in EMI/ EMC test with the same component and same board for led driver module only (Because it was the only module that was available)

Test procedure:

1. Design the PCB layout with EMI/EMC Considerations
2. Go the EMC lab and perform the DV test with the led taken as real load
3. Get the report
4. Get the old pcb layout and disconnect all module in it except for the LED Module only we want to compare
5. Go the Lab and test it with the conditions as the other board with the led taken as real load
6. Get the report
7. Compare the results

Layout tips:

1. DON'T MAKE A POLYGON ON THE Input of the board after the varistor choose the width the adequate with the current flowing through it



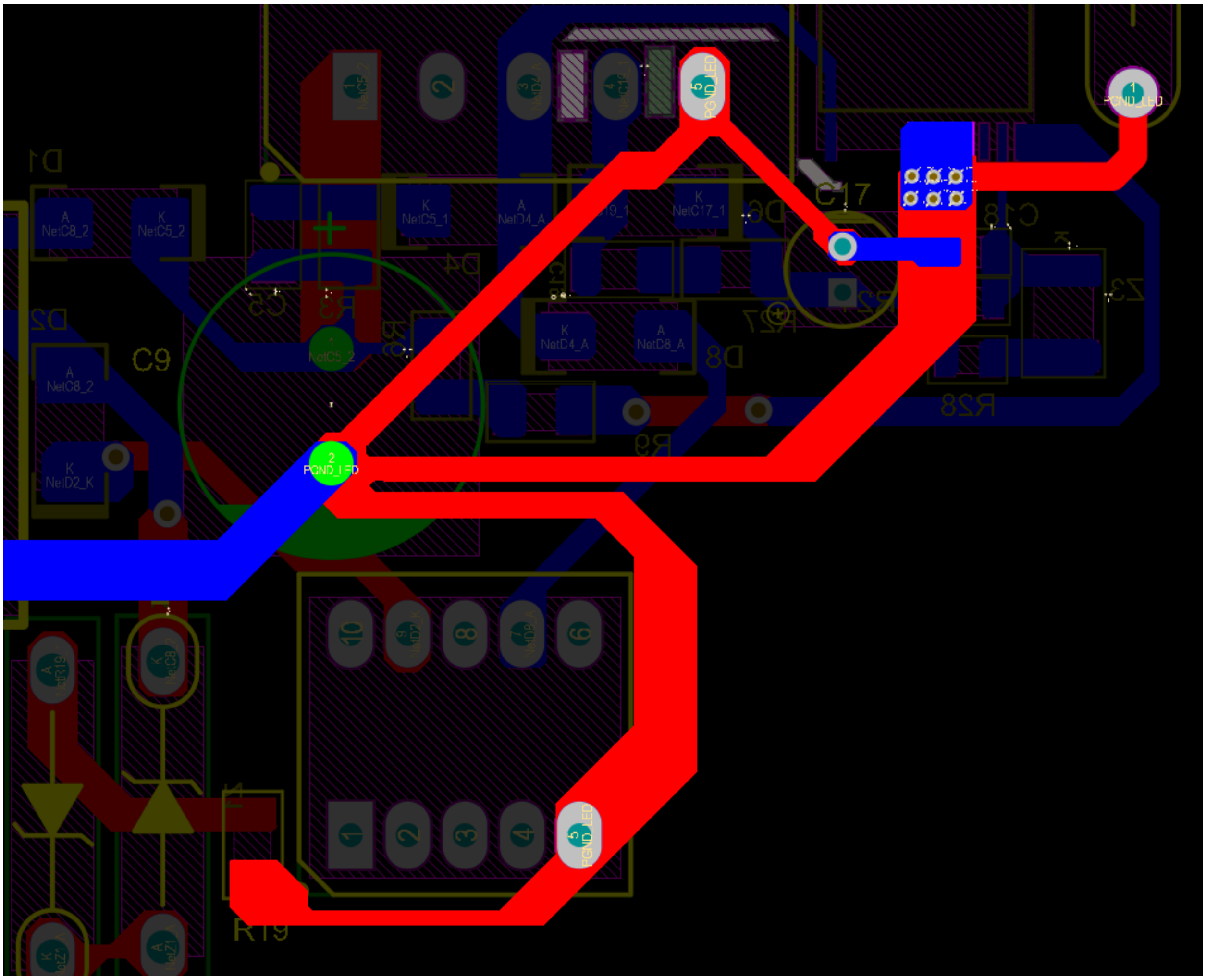
Why?

When we are taking about Immunity test like surge we are taking about +-2 KV high Energy Pulses that can be easily coupled to the rest surpass the varistor as very fast rise time its 8/20 Micro seconds wave-form

This very fast edge will try to find a low impedance path that can be a parasitic capacitor of the increased copper are to the input line and may be provide a lower impedance than the varistor making the varistor surge absorption is worse and even not functioning

As for EFT The rise time is 1ns MAKING THE PARASTICS CAPACITORS A REAL DEAL

Branching:



One of the most underestimated PCB Consideration though it's the major source of radiation and EMC issues and system errors

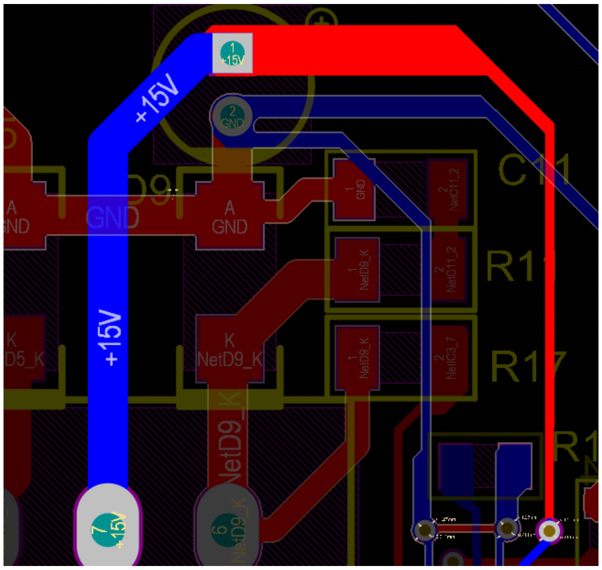
Branching insure the voltage seen by every component tied to it is theoretically is the same voltage potential as the gnd of capacitor which also should be ideal zero insuring that the correct functioning of the circuit tied to ground by making the same potential

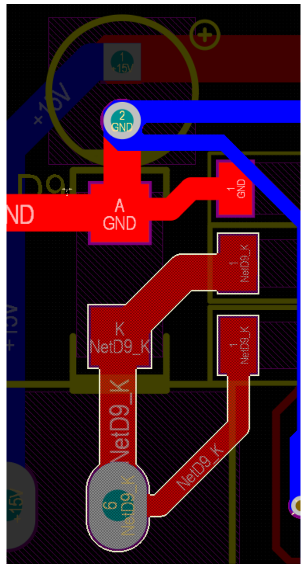
It's the most critical pcb layout consideration that must be take account

Why it is the most source of the radiation?

Because any voltage drop in the GND is considered as V-common Mode that is driving any cable tied to the PCB board to be an effective antenna causing radiations (Common-Mode rediation)

Separation high DI/DT of unrectified loops in the secondary to the rest of circuit.





IF the output of the transformer is on **bottom layer** as above make the + terminal of the capacitor as a via and continue routing the top layer

Also, IF the GND of the transformer is on **Top layer** as above make the – terminal of the capacitor as a via and via and continue routing the top layer.

For this topology of routing you have separated the high DI/DT Current of the transformer By using the capacitor terminals as a via note that many well-respected products companies are following this kind of topology making it Verfied

Final PCB LAYOUT

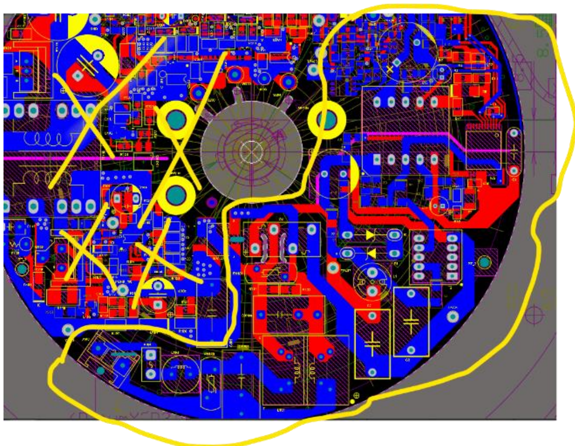


Figure 1 OLD LAY OUT

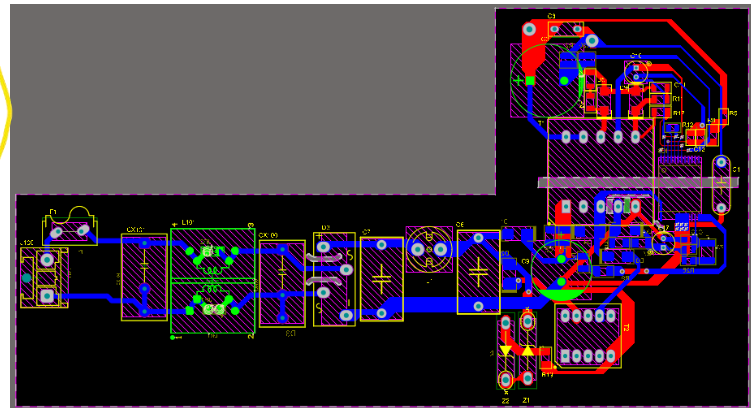


Figure 2NEW LAY OUY

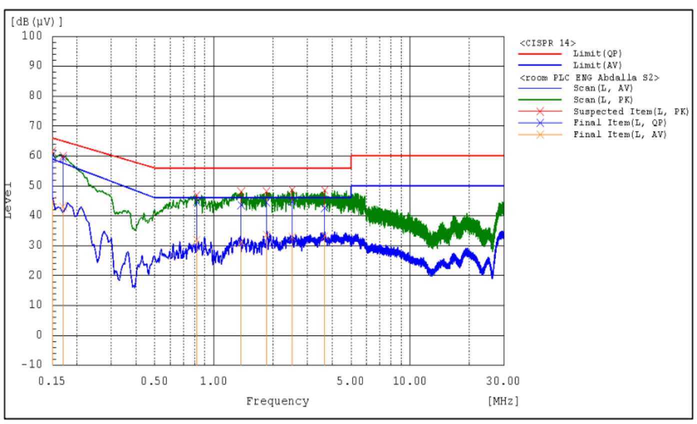
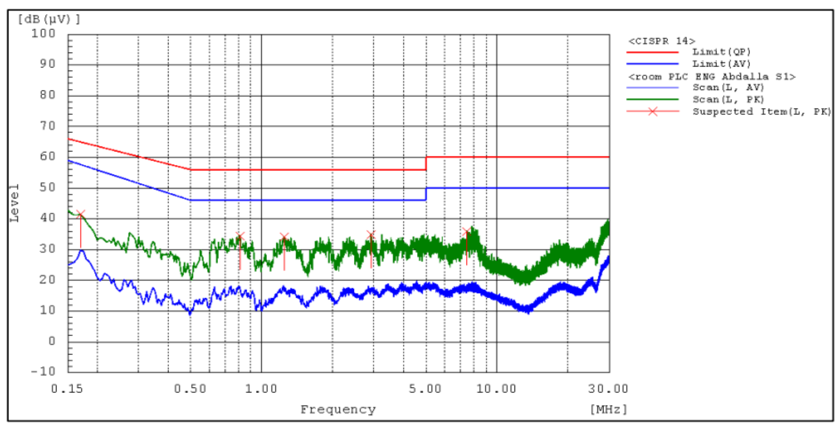


Fig 1: Disturbance Voltage spectrum of L-Terminal.



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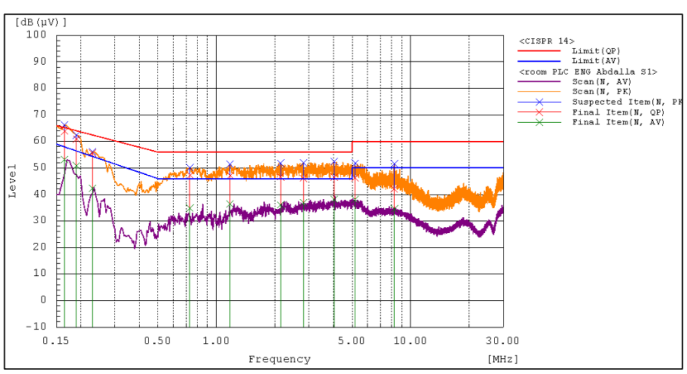

Fig 2: Disturbance Voltage spectrum of N-Terminal.

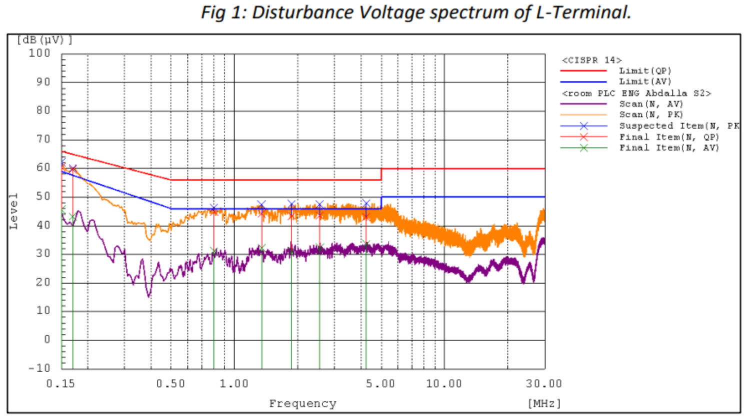

Fig 2: Disturbance Voltage spectrum of N-Terminal.

Final DATA FOR ENHANCED LAYOUT

Final Data

Frequency [MHz]	Line	Reading QP [dB(µV)]	Reading AV [dB(µV)]	Factor [dB]	Level QP [dB(µV)]	Level AV [dB(µV)]	Limit QP [dB(µV)]	Limit AV [dB(µV)]	Margin QP [dB]	Margin AV [dB]
0.150	L	50.3	34.9	10.2	60.5	45.1	66.0	59.0	5.5	13.9
0.170	L	49.1	32.9	10.2	59.3	43.1	65.0	57.6	5.7	14.5
0.815	L	34.6	21.7	10.2	44.8	31.9	56.0	46.0	11.2	14.1
1.375	L	33.3	21.0	10.3	43.6	31.3	56.0	46.0	12.4	14.7
1.850	L	34.3	23.3	10.3	44.6	33.6	56.0	46.0	11.4	12.4
2.500	L	34.4	22.6	10.3	44.7	32.9	56.0	46.0	11.3	13.1
3.665	L	32.3	22.9	10.3	42.6	33.2	56.0	46.0	13.4	12.8
0.150	N	50.3	35.4	10.2	60.5	45.6	66.0	59.0	5.5	13.4
0.170	N	49.6	33.1	10.2	59.8	43.3	65.0	57.6	5.2	14.3
0.800	N	34.7	21.0	10.2	44.9	31.2	56.0	46.0	11.1	14.8
1.345	N	33.9	21.9	10.3	44.2	32.2	56.0	46.0	11.8	13.8
1.870	N	33.1	20.9	10.3	43.4	31.2	56.0	46.0	12.6	14.8
2.545	N	33.3	22.2	10.3	43.6	32.5	56.0	46.0	12.4	13.5
4.230	N	32.9	23.3	10.3	43.2	33.6	56.0	46.0	12.8	12.4

Final DATA FOR OLD LAYOUT

Final Data

Frequency [MHz]	Line	Reading QP [dB(µV)]	Reading AV [dB(µV)]	Factor [dB]	Level QP [dB(µV)]	Level AV [dB(µV)]	Limit QP [dB(µV)]	Limit AV [dB(µV)]	Margin QP [dB]	Margin AV [dB]
0.165	N	53.7	43.0	10.2	63.9	53.2	65.2	58.0	1.3	4.8
0.190	N	50.7	40.7	10.2	60.9	50.9	64.0	56.4	3.1	5.5
0.230	N	45.4	32.3	10.2	55.6	42.5	62.4	54.4	6.8	11.9
0.730	N	37.1	24.7	10.2	47.3	34.9	56.0	46.0	8.7	11.1
1.175	N	36.9	26.2	10.3	47.2	36.5	56.0	46.0	8.8	9.5
2.140	N	36.4	25.8	10.3	46.7	36.1	56.0	46.0	9.3	9.9
2.805	N	35.9	26.7	10.3	46.2	37.0	56.0	46.0	9.8	9.0
4.040	N	36.2	27.9	10.3	46.5	38.2	56.0	46.0	9.5	7.8
5.175	N	36.0	27.9	10.3	46.3	38.2	60.0	50.0	13.7	11.8
8.250	N	31.8	24.3	10.4	42.2	34.7	60.0	50.0	17.8	15.3

Note that: the lowest margin in old layout is 1.3dB the enhanced layout is 5.5 dB enhancing by 323%

thanks